penev_B — formula report

497 inline formulas (section omitted; all rendered), 62 display equations, 1 tables, 17 unrecovered image regions.

Display equations

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>penev_B_ EQ0037 (5.19)</td><td>010</td><td>0.287
C|+6</td><td><code>\begin{aligned} A^{m}(\mathbf{x})= & {\left[\begin{array}{ll} p^{l}(\mathbf{x}) & P^{M}(\mathbf{x}) \end{array}\right]\left[\begin{array}{cc} \left(q^{-1} \right) r^{m} & \mathbf{0} \\ 0 & 0 \end{array} \right]+} \\ & +\bar{z}\left[\begin{array}{ll} p^{l}(\mathbf{x}) & P^{M}(\mathbf{x}) \end{array}\right] \left(\left[\begin{array}{c} -u^{T} \\ 1 \end{array} \right]\left[\begin{array}{ll} -u & 1 \end{array}\right] \right)_{1}^{m}= \\ = & {\left[\begin{array}{ll} p^{l}(\mathbf{x})\left(q^{-1}\right) I^{m} & 0 \end{array} \right]+\tilde{z}\left(P^{M}(\mathbf{x})-p^{l}(\mathbf{x}) u^{T}\right)\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right]=} \\ = & {\left[\begin{array}{ll} a^{m}(\mathbf{x}) & \mathbf{0} \end{array}\right]+ \bar{z} A(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right] } \end{aligned}</code></td><td>$$\begin{aligned} A^{m}(\mathbf{x})= & {\left[\begin{array}{ll} p^{l}(\mathbf{x}) & P^{M}(\mathbf{x}) \end{array}\right]\left[\begin{array}{cc} \left(q^{-1} \right) r^{m} & \mathbf{0} \\ 0 & 0 \end{array} \right]+} \\ & +\bar{z}\left[\begin{array}{ll} p^{l}(\mathbf{x}) & P^{M}(\mathbf{x}) \end{array}\right] \left(\left[\begin{array}{c} -u^{T} \\ 1 \end{array} \right]\left[\begin{array}{ll} -u & 1 \end{array}\right] \right)_{1}^{m}= \\ = & {\left[\begin{array}{ll} p^{l}(\mathbf{x})\left(q^{-1}\right) I^{m} & 0 \end{array} \right]+\tilde{z}\left(P^{M}(\mathbf{x})-p^{l}(\mathbf{x}) u^{T}\right)\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right]=} \\ = & {\left[\begin{array}{ll} a^{m}(\mathbf{x}) & \mathbf{0} \end{array}\right]+ \bar{z} A(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right] } \end{aligned}$$</td><td>$$\begin{aligned} A^{m}(\mathbf{x}) = & \left[\begin{array}{ll} p^{l}(\mathbf{x}) & P^{M}(\mathbf{x}) \end{array}\right]\left[\begin{array}{cc} (q^{-1})^{m}_{l} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{array}\right]+ \\ & +\bar{z}\left[\begin{array}{ll} p^{l}(\mathbf{x}) & P^{M}(\mathbf{x}) \end{array}\right]\left(\left[\begin{array}{c} -u^{T} \\ 1 \end{array}\right]\left[\begin{array}{ll} -u & 1 \end{array}\right]\right)^{m}_{l}= \\ = & \left[\begin{array}{ll} p^{l}(\mathbf{x})(q^{-1})^{m}_{l} & \mathbf{0} \end{array}\right]+\bar{z}(P^{M}(\mathbf{x})-p^{l}(\mathbf{x})u^{T}_{l})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right]= \\ = & \left[\begin{array}{ll} a^{m}(\mathbf{x}) & \mathbf{0} \end{array}\right]+\bar{z}A(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right] \qquad (5 \end{aligned}$$</td></tr>
<tr><td>penev_B_ EQ0005 (4.62)</td><td>001</td><td>0.325
N|+0</td><td><code>\mathrm{p}_{l}=\sum_{\mathrm{n}=1}^{|\mathcal{M}|} \mathrm{Q}^{-1}{ }_{\mathrm{ln}} \mathrm{q}_{\mathrm{n}} .</code></td><td>$$\mathrm{p}_{l}=\sum_{\mathrm{n}=1}^{|\mathcal{M}|} \mathrm{Q}^{-1}{ }_{\mathrm{ln}} \mathrm{q}_{\mathrm{n}} .$$</td><td>$$\mathbf{p}_{l}=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{}_{ln}\mathbf{q}_{n}.$$</td></tr>
<tr><td>penev_B_ EQ0016 (4.73)</td><td>003</td><td>0.397
W|+0</td><td><code>\begin{aligned} & O^{r e x}(\mathbf{x})=\sum_{i=1}^{| \mathcal{M}|} \mathcal{O}_{l} P_{l}(\mathbf{x}) \\ & \phi^{r e c}(\mathbf{x})=\sum_{l=1}^{|\mathcal{M}|} \mathcal{O}_{l} K_{l}^{-1}(\mathbf{x}) \end{aligned}</code></td><td>$$\begin{aligned} & O^{r e x}(\mathbf{x})=\sum_{i=1}^{| \mathcal{M}|} \mathcal{O}_{l} P_{l}(\mathbf{x}) \\ & \phi^{r e c}(\mathbf{x})=\sum_{l=1}^{|\mathcal{M}|} \mathcal{O}_{l} K_{l}^{-1}(\mathbf{x}) \end{aligned}$$</td><td>$$\begin{aligned} O^{rec}(\mathbf{x}) &= \sum_{l=1}^{|\mathcal{M}|} \mathcal{O}_{l}P_{l}(\mathbf{x}) \\ \phi^{rec}(\mathbf{x}) &= \sum_{l=1}^{|\mathcal{M}|} \mathcal{O}_{l}K_{l}^{-1}(\mathbf{x}). \end{aligned}$$</td></tr>
<tr><td>penev_B_ EQ0033 (5.15)</td><td>010</td><td>0.412
W|+1</td><td><code>\left[\begin{array}{cc} -u & 1 \end{array}\right] \equiv \left[\begin{array}{ll} -y w^{-1} & 1 \end{array}\right]= \left[\begin{array}{ll} -p q^{-1} & 1 \end{array}\right]</code></td><td>$$\left[\begin{array}{cc} -u & 1 \end{array}\right] \equiv \left[\begin{array}{ll} -y w^{-1} & 1 \end{array}\right]= \left[\begin{array}{ll} -p q^{-1} & 1 \end{array}\right]$$</td><td>$$\left[\begin{array}{cc} -u & 1 \end{array}\right] \equiv \left[\begin{array}{ll} -y w^{-1} & 1 \end{array}\right]= \left[\begin{array}{ll} -p q^{-1} & \mathbf{1} \end{array}\right]$$</td></tr>
<tr><td>penev_B_ EQ0050 (5.32)</td><td>011</td><td>0.503
C|+3</td><td><code>\bar{z}^{1}=P^{M}\left(\mathbf{x}_{M}\right)-a^{m}\left(\mathbf{x}_{M}\right) P^{M}\left(\mathbf{x}_{m}\right)= \mathcal{D}_{P^{M}}\left(\mathbf{x}_{M}\right) .</code></td><td>$$\bar{z}^{1}=P^{M}\left(\mathbf{x}_{M}\right)-a^{m}\left(\mathbf{x}_{M}\right) P^{M}\left(\mathbf{x}_{m}\right)= \mathcal{D}_{P^{M}}\left(\mathbf{x}_{M}\right) .$$</td><td>$$\bar{z}^{\;1}=P^{M}(\mathbf{x}_{M})-a^{m}(\mathbf{x}_{M})P^{M}(\mathbf{x}_{m})=\mathcal{D}_{P^{M}}(\mathbf{x}_{M}).$$</td></tr>
<tr><td>penev_B_ EQ0010 (4.67)</td><td>001</td><td>0.503
W|-1</td><td><code>O^{r e n}(\mathbf{x})=\sum_{m=1}^{|\mathcal{M}|} O_{m} a_{m}(\mathbf{x})</code></td><td>$$O^{r e n}(\mathbf{x})=\sum_{m=1}^{|\mathcal{M}|} O_{m} a_{m}(\mathbf{x})$$</td><td>$$O^{rec}(\mathbf{x})=\sum_{m=1}^{|\mathcal{M}|} O_{m}a_{m}(\mathbf{x})$$</td></tr>
<tr><td>penev_B_EQ0027 (5.9)</td><td>010</td><td>0.518
W|+1</td><td><code>W=\left[\begin{array}{cc} w & x \\ y & z \end{array} \right]=\left[\begin{array}{cc} 1_{n^{\prime}} & 0 \\ y w^{-1} & 1_{n^{\prime \prime}} \end{array}\right]\left[\begin{array}{cc} w & x \\ 0 & \tilde{z}^{-1} \end{array} \right]</code></td><td>$$W=\left[\begin{array}{cc} w & x \\ y & z \end{array} \right]=\left[\begin{array}{cc} 1_{n^{\prime}} & 0 \\ y w^{-1} & 1_{n^{\prime \prime}} \end{array}\right]\left[\begin{array}{cc} w & x \\ 0 & \tilde{z}^{-1} \end{array} \right]$$</td><td>$$W=\left[\begin{array}{cc} w & x \\ y & z \end{array}\right]=\left[\begin{array}{cc} 1_{n'} & \mathbf{0} \\ yw^{-1} & 1_{n''} \end{array}\right]\left[\begin{array}{cc} w & x \\ \mathbf{0} & \bar{z}^{-1} \end{array}\right]$$</td></tr>
</table>

Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>penev_B_ EQ0029 (5.11)</td><td>010</td><td>■0.537 ● C|+11</td><td><code>\begin{aligned} W^{1} & =\left[\begin{array}{cc} w^{1} & -w^{1} x \tilde{z} \\ 0 & \tilde{z} \end{array} \right]\left[\begin{array}{cc} 1 & 0 \\ -y w^{-1} & 1 \end{array}\right]= \\ & =\left[\begin{array}{cc} w^{-1}+w^{-1} x \bar{z} y w^{-1} & -w^{-1} x \bar{z} \\ -\tilde{z} y w^{-1} & \bar{z} \end{array}\right]= \\ & = \left[\begin{array}{cc} w^{-1} & 0 \\ 0 & 0 \end{array} \right]+\left[\begin{array}{cc} w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array}\right] . \end{aligned}</code></td><td>$\begin{aligned} W^{1} & =\left[\begin{array}{cc} w^{1} & -w^{1} x \tilde{z} \\ 0 & \tilde{z} \end{array} \right]\left[\begin{array}{cc} 1 & 0 \\ -y w^{-1} & 1 \end{array}\right]= \\ & =\left[\begin{array}{cc} w^{-1}+w^{-1} x \bar{z} y w^{-1} & -w^{-1} x \bar{z} \\ -\tilde{z} y w^{-1} & \bar{z} \end{array}\right]= \\ & = \left[\begin{array}{cc} w^{-1} & 0 \\ 0 & 0 \end{array} \right]+\left[\begin{array}{cc} w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array}\right] . \end{aligned}$</td><td>$\begin{aligned} W^{1} & =\left[\begin{array}{cc} w^{1} & -w^{1} x \tilde{z} \\ 0 & \tilde{z} \end{array} \right]\left[\begin{array}{cc} 1 & 0 \\ -y w^{-1} & 1 \end{array}\right]= \\ & =\left[\begin{array}{cc} w^{-1}+w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array}\right]= \\ & = \left[\begin{array}{cc} w^{-1} & 0 \\ 0 & 0 \end{array} \right]+\left[\begin{array}{cc} w^{-1} x \tilde{z} y w^{-1} & -w^{-1} x \tilde{z} \\ -\tilde{z} y w^{-1} & \tilde{z} \end{array}\right] . \end{aligned}$</td></tr>
<tr><td>penev_B_EQ0024 (5.6)</td><td>009</td><td>■0.542 ● C|+4</td><td><code>\mathbf{K}_{j}^{(n)}=\sum_{\mathbf{f} \in \Delta_{j}} \Psi_{\mathbf{f}}\left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{n}\left(\Psi_{\mathbf{f}},\right) .</code></td><td>$\mathbf{K}_{j}^{(n)}=\sum_{\mathbf{f} \in \Delta_{j}} \Psi_{\mathbf{f}}\left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{n}\left(\Psi_{\mathbf{f}},\right) .$</td><td>$\mathbf{K}_{j}^{(n)}=\sum_{\mathbf{f} \in \Delta_{j}} \Psi_{\mathbf{f}}\left(\frac{1}{\sigma_{\mathbf{f}}} \right)^{n}\left(\Psi_{\mathbf{f}},\ \right).$</td></tr>
<tr><td>penev_B_ EQ0013 (4.70)</td><td>001</td><td>■0.665 ● W|+1</td><td><code>\phi^{r e c}(\mathrm{x})=\sum_{m=1}^{|\mathcal{M}|} O_{m} \sum_{n=1}^{|\mathcal{M}|} \mathrm{Q}_{m n}^{-1} K^{(-1)}{ }_{n}(\mathrm{x})</code></td><td>$\phi^{r e c}(\mathrm{x})=\sum_{m=1}^{|\mathcal{M}|} O_{m} \sum_{n=1}^{|\mathcal{M}|} \mathrm{Q}_{m n}^{-1} K^{(-1)}{ }_{n}(\mathrm{x})$</td><td>$\phi^{rec}(\mathbf{x})=\sum_{m=1}^{|\mathcal{M}|} O_{m} \sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{}_{mn} K^{(-1)}{}_{n}(\mathbf{x})$</td></tr>
<tr><td>penev_B_EQ0026 (5.8)</td><td>009</td><td>■0.668 ● C|+4</td><td><code>\mathbf{O}_{j}^{*} \mathbf{O}_{j}=\mathbf{P}_{j} \neq \mathbf{1} .</code></td><td>$\mathbf{O}_{j}^{*} \mathbf{O}_{j}=\mathbf{P}_{j} \neq \mathbf{1} .$</td><td>$\mathbf{O}_{j}^{*} \mathbf{O}_{j}=\mathbf{P}_{j} \neq \mathbf{1} .$</td></tr>
<tr><td>penev_B_ EQ0053 (5.35)</td><td>011</td><td>■0.670 ● W|+2</td><td><code>\begin{aligned} \Delta O^{r e c}(\mathbf{x}) & =\Delta A^{m}(\mathbf{x}) O\left(\mathbf{x}_{m}\right)= \\ & =A^{M}(\mathbf{x})\left[-a^{m}\left(\mathbf{x}_{M} \right) \quad 1\right]\left[\begin{array}{l} O \left(\mathbf{x}_{m}\right) \\ O\left(\mathbf{x}_{M} \right) \end{array}\right]= \\ & =A^{M}(\mathbf{x}) \mathcal{D}_{O}\left(\mathbf{x}_{M}\right) \end{aligned}</code></td><td>$\begin{aligned} \Delta O^{r e c}(\mathbf{x}) & =\Delta A^{m}(\mathbf{x}) O\left(\mathbf{x}_{m}\right)= \\ & =A^{M}(\mathbf{x})\left[-a^{m}\left(\mathbf{x}_{M} \right) \quad 1\right]\left[\begin{array}{l} O \left(\mathbf{x}_{m}\right) \\ O\left(\mathbf{x}_{M} \right) \end{array}\right]= \\ & =A^{M}(\mathbf{x}) \mathcal{D}_{O}\left(\mathbf{x}_{M}\right) \end{aligned}$</td><td>$\begin{aligned} \Delta O^{rec}(\mathbf{x}) & =\Delta A^{m}(\mathbf{x}) O(\mathbf{x}_{m})= \\ & =A^{M}(\mathbf{x})\left[-a^{m}(\mathbf{x}_{M}) \quad 1\right]\left[\begin{array}{l} O(\mathbf{x}_{m}) \\ O(\mathbf{x}_{M}) \end{array}\right]= \\ & =A^{M}(\mathbf{x}) \mathcal{D}_{O}(\mathbf{x}_{M}). \end{aligned}$</td></tr>
<tr><td>penev_B_ EQ0018 (4.75)</td><td>004</td><td>■0.672 ● W|-1</td><td><code>-\tau \frac{d}{d t} \mathcal{O}(\mathbf{x})=\mathrm{K} \phi-\mathbf{P} G(\mathcal{O})</code></td><td>$-\tau \frac{d}{d t} \mathcal{O}(\mathbf{x})=\mathrm{K} \phi-\mathbf{P} G(\mathcal{O})$</td><td>$-\tau \frac{d}{dt} \mathcal{O}(\mathbf{x})=\mathrm{K} \phi-\mathbf{P} G(\mathcal{O})$</td></tr>
<tr><td>penev_B_ EQ0034 (5.16)</td><td>010</td><td>■0.682 ● W|+1</td><td><code>\mathrm{Q}^{-1}=\left[\begin{array}{cc} q^{-1} & 0 \\ 0 & 0 \end{array}\right]+\bar{z}\left[\begin{array}{c} -u^{T} \\ 1 \end{array}\right]\left[\begin{array}{ll} -u & 1 \end{array}\right] .</code></td><td>$\mathrm{Q}^{-1}=\left[\begin{array}{cc} q^{-1} & 0 \\ 0 & 0 \end{array}\right]+\bar{z}\left[\begin{array}{c} -u^{T} \\ 1 \end{array}\right]\left[\begin{array}{ll} -u & 1 \end{array}\right] .$</td><td>$\mathbf{Q}^{-1}=\left[\begin{array}{cc} q^{-1} & 0 \\ 0 & 0 \end{array}\right]+\tilde{z}\left[\begin{array}{c} -u^{T} \\ 1 \end{array}\right]\left[\begin{array}{ll} -u & 1 \end{array}\right].$</td></tr>
<tr><td>penev_B_ EQ0046 (5.28)</td><td>011</td><td>■0.695 ● N|+2</td><td><code>A(\mathbf{x})=P^{M}(\mathbf{x})-\mathcal{R}_{P^{M}}^{M-1}(\mathbf{x}) .</code></td><td>$A(\mathbf{x})=P^{M}(\mathbf{x})-\mathcal{R}_{P^{M}}^{M-1}(\mathbf{x}) .$</td><td>$A(\mathbf{x})=P^{M}(\mathbf{x})-\mathcal{R}_{P^{M}}^{M-1}(\mathbf{x}).$</td></tr>
<tr><td>penev_B_ EQ0002 (4.59)</td><td>001</td><td>■0.709 ● C|+3</td><td><code>O^{\text {rec }}(\mathbf{x})=\left(\hat{\mathbf{x}}, \mathbf{O}^{*} \sum_{m=1}^{|\mathcal{M}|} O_{m} \mathbf{p}_{m}\right)=\sum_{m=1}^{|\mathcal{M}|} O_{m} \left(\hat{\mathbf{x}}, \mathrm{O}^{*} \mathbf{p}_{m} \right)=\sum_{m=1}^{\mid \mathcal{M}} O_{m} a_{m}(\mathbf{x})</code></td><td>$O^{\text {rec }}(\mathbf{x})=\left(\hat{\mathbf{x}}, \mathbf{O}^{*} \sum_{m=1}^{|\mathcal{M}|} O_{m} \mathbf{p}_{m}\right)=\sum_{m=1}^{|\mathcal{M}|} O_{m} \left(\hat{\mathbf{x}}, \mathrm{O}^{*} \mathbf{p}_{m} \right)=\sum_{m=1}^{\mid \mathcal{M}} O_{m} a_{m}(\mathbf{x})$</td><td>$O^{rec}(\mathbf{x})=(\hat{\mathbf{x}}, \mathbf{O}^{*} \sum_{m=1}^{|\mathcal{M}|} O_{m} \mathbf{p}_{m})=\sum_{m=1}^{|\mathcal{M}|} O_{m} (\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{p}_{m})=\sum_{m=1}^{|\mathcal{M}} O_{m} a_{m}(\mathbf{x})$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>penev_B_ EQ0009 (4.66)</td><td>001</td><td>0.710
W|+0</td><td><code>a_{m}(\mathbf{x})=\left\langle\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{p}_{m}\right\rangle=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}_{m n}^{-1}\left(\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{q}_{n}\right)=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}_{m n}^{-1} P_{n}(\mathbf{x}) .</code></td><td>$a_{m}(\mathbf{x})=\left\langle\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{p}_{m}\right\rangle=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}_{m n}^{-1}\left(\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{q}_{n}\right)=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}_{m n}^{-1} P_{n}(\mathbf{x}) .$</td><td>$a_m(\mathbf{x}) = \langle\hat{\mathbf{x}}, \mathbf{O}^*\mathbf{p}_m\rangle = \sum_{n=1}^{|\mathcal{M}|}\mathbf{Q}^{-1}{}_{mn}\langle\hat{\mathbf{x}}, \mathbf{O}^*\mathbf{q}_n\rangle = \sum_{n=1}^{|\mathcal{M}|}\mathbf{Q}^{-1}{}_{mn}P_n(\mathbf{x}).$</td></tr>
<tr><td>penev_B_ EQ0062 (5.43)</td><td>012</td><td>0.779
S|+2</td><td><code>\Delta v^{r e c}(t)=A^{M}(t) \times 1</code></td><td>$\Delta v^{r e c}(t)=A^{M}(t) \times 1$</td><td>$\Delta v^{rec}(t) = A^M(t) \times 1$</td></tr>
<tr><td>penev_B_ EQ0044 (5.26)</td><td>010</td><td>0.782
N|+0</td><td><code>\mathcal{R}_{V}^{|\mathcal{M}|}(\mathbf{x})=A^{m}(\mathbf{x}) V\left(\mathbf{x}_{\boldsymbol{m}}\right) .</code></td><td>$\mathcal{R}_{V}^{|\mathcal{M}|}(\mathbf{x})=A^{m}(\mathbf{x}) V\left(\mathbf{x}_{\boldsymbol{m}}\right) .$</td><td>$\mathcal{R}_V^{|\mathcal{M}|}(\mathbf{x}) = A^m(\mathbf{x})V(\mathbf{x}_m).$</td></tr>
<tr><td>penev_B_ EQ0061 (5.42)</td><td>011</td><td>0.825
N|+0</td><td><code>\Delta f^{r e c}\left(\mathbf{x}_{M}\right)=e_{\mathbf{n}}=1 \times \mathcal{D}_{f}\left(\mathbf{x}_{M}\right) .</code></td><td>$\Delta f^{r e c}\left(\mathbf{x}_{M}\right)=e_{\mathbf{n}}=1 \times \mathcal{D}_{f}\left(\mathbf{x}_{M}\right) .$</td><td>$\Delta f^{rec}(\mathbf{x}_M) = e_n = 1 \times \mathcal{D}_f(\mathbf{x}_M).$</td></tr>
<tr><td>penev_B_ EQ0059 (5.40)</td><td>011</td><td>0.868
W|+1</td><td><code>Q_{m l}=\left.E\left\{f_{n-l} f_{n-m}\right\}\right|_{m, l=0} ^{M-1}</code></td><td>$Q_{m l}=\left.E\left\{f_{n-l} f_{n-m}\right\}\right|_{m, l=0} ^{M-1}$</td><td>$Q_{ml} = E\{f_{n-l}f_{n-m}\}\big|_{m,l=0}^{M-1}$</td></tr>
<tr><td>penev_B_ EQ0007 (4.64)</td><td>001</td><td>0.887
N|+0</td><td><code>P_{m}(\mathrm{x}) \equiv P_{\mathrm{x}_{\mathrm{m}}}(\mathrm{x}) .</code></td><td>$P_{m}(\mathrm{x}) \equiv P_{\mathrm{x}_{\mathrm{m}}}(\mathrm{x}) .$</td><td>$P_m(\mathbf{x}) \equiv P_{\mathbf{x}_m}(\mathbf{x}).$</td></tr>
<tr><td>penev_B_ EQ0036 (5.18)</td><td>010</td><td>0.923
S|+1</td><td><code>A^{m}(\mathbf{x})=P^{l}(\mathbf{x})\left(\mathbf{Q}^{-1}\right)_{l}^{m}</code></td><td>$A^{m}(\mathbf{x})=P^{l}(\mathbf{x})\left(\mathbf{Q}^{-1}\right)_{l}^{m}$</td><td>$A^m(\mathbf{x}) = P^l(\mathbf{x})(\mathbf{Q}^{-1})_l^m$</td></tr>
<tr><td>penev_B_ EQ0015 (4.72)</td><td>003</td><td>0.924
N|+0</td><td><code>\mathcal{O}_{l} \equiv \mathbf{Q}^{-1}{ }_{l m} O_{m} .</code></td><td>$\mathcal{O}_{l} \equiv \mathbf{Q}^{-1}{ }_{l m} O_{m} .$</td><td>$\mathcal{O}_l \equiv \mathbf{Q}^{-1}{}_{lm}O_m.$</td></tr>
<tr><td>penev_B_ EQ0045 (5.27)</td><td>011</td><td>0.925
N|-1</td><td><code>O^{r e c}(\mathbf{x})=\mathcal{R}_{O}^{M}(\mathbf{x})</code></td><td>$O^{r e c}(\mathbf{x})=\mathcal{R}_{O}^{M}(\mathbf{x})$</td><td>$O^{rec}(\mathbf{x}) = \mathcal{R}_O^M(\mathbf{x})$</td></tr>
<tr><td>penev_B_EQ0022 (5.4)</td><td>006</td><td>0.928
W|+0</td><td><code>N\langle S\rangle(N, \alpha)=\left\|\Phi_{N}^{*+} \phi(\alpha)\right\|_{S}^{2} .</code></td><td>$N\langle S\rangle(N, \alpha)=\left\|\Phi_{N}^{*+} \phi(\alpha)\right\|_{S}^{2} .$</td><td>$N\langle S\rangle(N,\alpha) = \|\Phi^{*+}_N\phi(\alpha)\|_S^2.$</td></tr>
<tr><td>penev_B_ EQ0030 (5.12)</td><td>010</td><td>0.929
W|+0</td><td><code>Q_{m l}=\left.P_{m}\left(\mathbf{x}_{l}\right)\right|_{l, m=1} ^{M} .</code></td><td>$Q_{m l}=\left.P_{m}\left(\mathbf{x}_{l}\right)\right|_{l, m=1} ^{M} .$</td><td>$Q_{ml} = P_m(\mathbf{x}_l)\big|_{l,m=1}^M.$</td></tr>
<tr><td>penev_B_ EQ0043 (5.25)</td><td>010</td><td>0.933
S|+1</td><td><code>A(\mathbf{x})=P^{M}(\mathbf{x})-a^{\mathbf{m}}(\mathbf{x}) P^{M}\left(\mathbf{x}_{\boldsymbol{m}}\right) .</code></td><td>$A(\mathbf{x})=P^{M}(\mathbf{x})-a^{\mathbf{m}}(\mathbf{x}) P^{M}\left(\mathbf{x}_{\boldsymbol{m}}\right) .$</td><td>$A(\mathbf{x}) = P^M(\mathbf{x}) - a^m(\mathbf{x})P^M(\mathbf{x}_m).$</td></tr>
<tr><td>penev_B_ EQ0054 (5.36)</td><td>011</td><td>0.941
N|+0</td><td><code>\Delta O^{r e c}(\mathbf{x})=\frac{\mathcal{D}_{P^{M}}(\mathbf{x})}{\mathcal{D}_{P^{M}}\left(\mathbf{x}_{M}\right)} \mathcal{D}_{O}\left(\mathbf{x}_{M}\right)</code></td><td>$\Delta O^{r e c}(\mathbf{x})=\frac{\mathcal{D}_{P^{M}}(\mathbf{x})}{\mathcal{D}_{P^{M}}\left(\mathbf{x}_{M}\right)} \mathcal{D}_{O}\left(\mathbf{x}_{M}\right)$</td><td>$\Delta O^{rec}(\mathbf{x}) = \frac{\mathcal{D}_{P^M}(\mathbf{x})}{\mathcal{D}_{P^M}(\mathbf{x}_M)}\mathcal{D}_O(\mathbf{x}_M).$</td></tr>
<tr><td>penev_B_ EQ0014 (4.71)</td><td>001</td><td>0.941
N|-1</td><td><code>O^{\text {err }}(\mathbf{x})=O(\mathbf{x})-O^{\text {rec }}(\mathbf{x}) .</code></td><td>$O^{\text {err }}(\mathbf{x})=O(\mathbf{x})-O^{\text {rec }}(\mathbf{x}) .$</td><td>$O^{err}(\mathbf{x}) = O(\mathbf{x}) - O^{rec}(\mathbf{x}).$</td></tr>
<tr><td>penev_B_ EQ0035 (5.17)</td><td>010</td><td>0.948
N|-2</td><td><code>O^{r e c}(\mathbf{x})=A^{m}(\mathbf{x}) O\left(\mathbf{x}_{m}\right)</code></td><td>$O^{r e c}(\mathbf{x})=A^{m}(\mathbf{x}) O\left(\mathbf{x}_{m}\right)$</td><td>$O^{rec}(\mathbf{x}) = A^m(\mathbf{x})O(\mathbf{x}_m)$</td></tr>
<tr><td>penev_B_EQ0020 (5.1)</td><td>005</td><td>0.950
N|+2</td><td><code>\alpha^{t}=\left(\theta^{t}, S^{t}, X^{t}, Y^{t}\right) .</code></td><td>$\alpha^{t}=\left(\theta^{t}, S^{t}, X^{t}, Y^{t}\right) .$</td><td>$\alpha^t = (\theta^t, S^t, X^t, Y^t).$</td></tr>
<tr><td>penev_B_ EQ0057 (5.38)</td><td>011</td><td>0.956
K|+0</td><td><code>\alpha^{m}=\left(q^{-1}\right)_{l}^{m} Q_{0}^{l}</code></td><td>$\alpha^{m}=\left(q^{-1}\right)_{l}^{m} Q_{0}^{l}$</td><td>$\alpha^m = (q^{-1})_l^m Q_0^l$</td></tr>
<tr><td>penev_B_ EQ0058 (5.39)</td><td>011</td><td>0.962
W|-2</td><td><code>q_{m l}=\left.Q_{m l}\right|_{m, l=1} ^{M-1}</code></td><td>$q_{m l}=\left.Q_{m l}\right|_{m, l=1} ^{M-1}$</td><td>$q_{ml} = Q_{ml}\big|_{m,l=1}^{M-1}$</td></tr>
<tr><td>penev_B_ EQ0019 (4.76)</td><td>004</td><td>0.963
N|+1</td><td><code>\mathbf{P O}=\mathbf{K} \phi .</code></td><td>$\mathbf{P O}=\mathbf{K} \phi .$</td><td>$\mathbf{P}\mathcal{O} = \mathbf{K}\phi.$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value
Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured</td></tr>
</table>

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>penev_B_ EQ0012 (4.69)</td><td>001</td><td>0.971 W|+2</td><td><code>a_{m}\left(\mathbf{x}_{l}\right)=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{ }_{m n} P_{n}\left(\mathbf{x}_{l}\right)=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{ }_{m n} \mathbf{Q}_{n l}=\delta_{m l} .</code></td><td>$a_{m}\left(\mathbf{x}_{l}\right)=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{ }_{m n} P_{n}\left(\mathbf{x}_{l}\right)=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{ }_{m n} \mathbf{Q}_{n l}=\delta_{m l} .$</td><td>$a_m(\mathbf{x}_l)=\sum_{n=1}^{|\mathcal{M}|}\mathbf{Q}^{-1}{}_{mn}P_n(\mathbf{x}_l)=\sum_{n=1}^{|\mathcal{M}|}\mathbf{Q}^{-1}{}_{mn}\mathbf{Q}_{nl}=\delta_{m,l}.$</td></tr>
<tr><td>penev_B_ EQ0028 (5.10)</td><td>010</td><td>0.979 N|+0</td><td><code>\tilde{z}^{1}=z \quad y w{ }^{1} x</code></td><td>$\tilde{z}^{1}=z \quad y w{ }^{1} x$</td><td>$\tilde{z}\ {}^{1}=z \quad yw\ {}^{1}x$</td></tr>
<tr><td>penev_B_ EQ0017 (4.74)</td><td>003</td><td>0.981 W|+2</td><td><code>\sum_{l=1}^{|\mathcal{M}|} \mathrm{Q}_{m l} \mathcal{O}_{l}=O_{m}</code></td><td>$\sum_{l=1}^{|\mathcal{M}|} \mathrm{Q}_{m l} \mathcal{O}_{l}=O_{m}$</td><td>$\sum_{l=1}^{|\mathcal{M}|}\mathbf{Q}_{ml}\mathcal{O}_l=O_m$</td></tr>
<tr><td>penev_B_ EQ0003 (4.60)</td><td>001</td><td>0.981 N|+0</td><td><code>a_{m}(\mathbf{x}) \equiv\left(\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{p}_{m}\right)</code></td><td>$a_{m}(\mathbf{x}) \equiv\left(\hat{\mathbf{x}}, \mathbf{O}^{*} \mathbf{p}_{m}\right)$</td><td>$a_m(\mathbf{x})\equiv(\hat{\mathbf{x}},\mathbf{O}^*\mathbf{p}_m).$</td></tr>
<tr><td>penev_B_ EQ0004 (4.61)</td><td>001</td><td>0.984 S|+0</td><td><code>\delta_{m, l}=\left\langle\mathbf{q}_{m}, \sum_{n=1}^{|\mathcal{M}|} \mathbf{B}_{l n} \mathbf{q}_{n}\right\rangle=\sum_{n=1}^{\mathcal{M} \mid} \mathbf{B}_{l n} \left\langle\mathbf{q}_{m}, \mathbf{q}_{n}\right\rangle=\sum_{n=1}^{|\mathcal{M}|} \mathbf{B}_{l n} \mathbf{Q}_{m n}</code></td><td>$\delta_{m, l}=\left\langle\mathbf{q}_{m}, \sum_{n=1}^{|\mathcal{M}|} \mathbf{B}_{l n} \mathbf{q}_{n}\right\rangle=\sum_{n=1}^{\mathcal{M} \mid} \mathbf{B}_{l n} \left\langle\mathbf{q}_{m}, \mathbf{q}_{n}\right\rangle=\sum_{n=1}^{|\mathcal{M}|} \mathbf{B}_{l n} \mathbf{Q}_{m n}$</td><td>$\delta_{m,l}=\langle\mathbf{q}_m,\sum_{n=1}^{|\mathcal{M}|}\mathbf{B}_{ln}\mathbf{q}_n\rangle=\sum_{n=1}^{\mathcal{M}|}\mathbf{B}_{ln}\langle\mathbf{q}_m,\mathbf{q}_n\rangle=\sum_{n=1}^{|\mathcal{M}|}\mathbf{B}_{ln}\mathbf{Q}_{mn}$</td></tr>
<tr><td>penev_B_ EQ0011 (4.68)</td><td>001</td><td>0.991 S|+0</td><td><code>a_{m}(\mathbf{x})=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{ }_{m n} P_{n}(\mathbf{x}) .</code></td><td>$a_{m}(\mathbf{x})=\sum_{n=1}^{|\mathcal{M}|} \mathbf{Q}^{-1}{ }_{m n} P_{n}(\mathbf{x}) .$</td><td>$a_m(\mathbf{x})=\sum_{n=1}^{|\mathcal{M}|}\mathbf{Q}^{-1}{}_{mn}P_n(\mathbf{x}).$</td></tr>
<tr><td>penev_B_ EQ0041 (5.23)</td><td>010</td><td>0.992 C|+3</td><td><code>\Delta A^{m}(\mathbf{x})=A^{M}(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right] .</code></td><td>$\Delta A^{m}(\mathbf{x})=A^{M}(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right] .$</td><td>$\Delta A^m(\mathbf{x})=A^M(\mathbf{x})\left[\begin{array}{ll}-u^m & 1\end{array}\right].$</td></tr>
<tr><td>penev_B_ EQ0060 (5.41)</td><td>011</td><td>0.995 N|+1</td><td><code>e_{n}=f_{n}-\hat{f}_{n}=\mathcal{D}_{f}\left(\mathbf{x}_{M}\right)</code></td><td>$e_{n}=f_{n}-\hat{f}_{n}=\mathcal{D}_{f}\left(\mathbf{x}_{M}\right)$</td><td>$e_n=f_n-\hat{f}_n=\mathcal{D}_f(\mathbf{x}_M)$</td></tr>
<tr><td>penev_B_ EQ0047 (5.29)</td><td>011</td><td>0.999 N|+2</td><td><code>\mathcal{D}_{V}(\mathbf{x})=V(\mathbf{x})-\mathcal{R}_{V}^{|\mathcal{M}|-1}(\mathbf{x})</code></td><td>$\mathcal{D}_{V}(\mathbf{x})=V(\mathbf{x})-\mathcal{R}_{V}^{|\mathcal{M}|-1}(\mathbf{x})$</td><td>$\mathcal{D}_V(\mathbf{x})=V(\mathbf{x})-\mathcal{R}_V^{|\mathcal{M}|-1}(\mathbf{x})$</td></tr>
<tr><td>penev_B_ EQ0006 (4.63)</td><td>001</td><td>0.999 N|+0</td><td><code>\mathbf{Q}_{n m}=\left\langle\mathbf{q}_{n}, \mathbf{q}_{m}\right\rangle=\left\langle\mathbf{O} \hat{\mathbf{x}}_{n}, \mathbf{q}_{m}\right\rangle=P_{m} \left(\mathbf{x}_{n}\right)</code></td><td>$\mathbf{Q}_{n m}=\left\langle\mathbf{q}_{n}, \mathbf{q}_{m}\right\rangle=\left\langle\mathbf{O} \hat{\mathbf{x}}_{n}, \mathbf{q}_{m}\right\rangle=P_{m} \left(\mathbf{x}_{n}\right)$</td><td>$\mathbf{Q}_{nm}=\langle\mathbf{q}_n,\mathbf{q}_m\rangle=\langle\mathbf{O}\hat{\mathbf{x}}_n,\mathbf{q}_m\rangle=P_m(\mathbf{x}_n)$</td></tr>
<tr><td>penev_B_ EQ0042 (5.24)</td><td>010</td><td>0.999 S|+0</td><td><code>u^{m}=p^{l}\left(\mathbf{x}_{M}\right)\left(q^{-1} \right)_{l}^{m}=a^{m}\left(\mathbf{x}_{M}\right) .</code></td><td>$u^{m}=p^{l}\left(\mathbf{x}_{M}\right)\left(q^{-1} \right)_{l}^{m}=a^{m}\left(\mathbf{x}_{M}\right) .$</td><td>$u^m=p^l(\mathbf{x}_M)(q^{-1})_l^m=a^m(\mathbf{x}_M).$</td></tr>
<tr><td>penev_B_ EQ0001 (4.58)</td><td>001</td><td>1.000 W|+1</td><td><code>s^{r e c}=s_{\|}=\sum_{m=1}^{|M|} O_{m} p_{m}</code></td><td>$s^{r e c}=s_{\|}=\sum_{m=1}^{|M|} O_{m} p_{m}$</td><td>$\mathbf{s}^{rec}=\mathbf{s}_{\|}=\sum_{m=1}^{|\mathcal{M}|}O_m\mathbf{p}_m.$</td></tr>
<tr><td>penev_B_ EQ0031 (5.13)</td><td>010</td><td>1.000 N|+0</td><td><code>p^{T}{ }_{l} \equiv y^{T}{ }_{l}=x_{l}=Q_{M l}= \left.P_{M}\left(\mathbf{x}_{l}\right)\right|_{l=1} ^{M-1} .</code></td><td>$p^{T}{ }_{l} \equiv y^{T}{ }_{l}=x_{l}=Q_{M l}= \left.P_{M}\left(\mathbf{x}_{l}\right)\right|_{l=1} ^{M-1} .$</td><td>$p^T{}_l\equiv y^T{}_l=x_l=Q_{Ml}=P_M(\mathbf{x}_l)|_{l=1}^{M-1}.$</td></tr>
<tr><td>penev_B_ EQ0008 (4.65)</td><td>001</td><td>1.000 N|+0</td><td><code>\mathbf{Q}=\left.\mathbf{P}\right|_{\mathcal{M}} .</code></td><td>$\mathbf{Q}=\left.\mathbf{P}\right|_{\mathcal{M}} .$</td><td>$\mathbf{Q}=\mathbf{P}|_{\mathcal{M}}.$</td></tr>
<tr><td>penev_B_EQ0021 (5.2)</td><td>005</td><td>1.000 N|+0</td><td><code>\phi^{t}=H\left(I^{t} ; \alpha^{t}\right)</code></td><td>$\phi^{t}=H\left(I^{t} ; \alpha^{t}\right)$</td><td>$\phi^t=H(I^t;\alpha^t)$</td></tr>
<tr><td>penev_B_EQ0023 (5.5)</td><td>007</td><td>1.000 N|+0</td><td><code>\alpha_{h}^{t}=\left(\theta^{t},(1+h) S^{t}, X^{t}, Y^{t} \right)</code></td><td>$\alpha_{h}^{t}=\left(\theta^{t},(1+h) S^{t}, X^{t}, Y^{t} \right)$</td><td>$\alpha_h^t=(\theta^t,(1+h)S^t,X^t,Y^t)$</td></tr>
<tr><td>penev_B_EQ0025 (5.7)</td><td>009</td><td>1.000 N|-1</td><td><code>O_{j}=\mathbf{K}_{j} \phi_{j}=\mathbf{K}_{j} \mathbf{P}_{j} \phi=\mathbf{K}_{j} \phi</code></td><td>$O_{j}=\mathbf{K}_{j} \phi_{j}=\mathbf{K}_{j} \mathbf{P}_{j} \phi=\mathbf{K}_{j} \phi$</td><td>$O_j=\mathbf{K}_j\phi_j=\mathbf{K}_j\mathbf{P}_j\phi=\mathbf{K}_j\phi$</td></tr>
<tr><td>penev_B_ EQ0032 (5.14)</td><td>010</td><td>1.000 N|+0</td><td><code>z=P_{M}\left(\mathbf{x}_{M}\right) .</code></td><td>$z=P_{M}\left(\mathbf{x}_{M}\right) .$</td><td>$z=P_M(\mathbf{x}_M).$</td></tr>
<tr><td>penev_B_ EQ0038 (5.20)</td><td>010</td><td>1.000 N|+2</td><td><code>A(\mathbf{x}) \equiv P^{M}(\mathbf{x})-p^{l}(\mathbf{x}) u^{T}{ }_{l}</code></td><td>$A(\mathbf{x}) \equiv P^{M}(\mathbf{x})-p^{l}(\mathbf{x}) u^{T}{ }_{l}$</td><td>$A(\mathbf{x})\equiv P^M(\mathbf{x})-p^l(\mathbf{x})u^T{}_l.$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
penev_B_ EQ0039 (5.21)	010	■ 1.000 ● N|+0	`\Delta A^{m}(\mathbf{x})=\bar{z} A(\mathbf{x})\left[ \begin{array}{ll} -u^{m} & 1 \end{array}\right]`	$\Delta A^{m}(\mathbf{x})=\bar{z} A(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right]$	$\Delta A^{m}(\mathbf{x})=\bar{z} A(\mathbf{x})\left[\begin{array}{ll} -u^{m} & 1 \end{array}\right]$
penev_B_ EQ0040 (5.22)	010	■ 1.000 ● N|+0	`A^{M}(\mathbf{x})=\bar{z} A(\mathbf{x})`	$A^{M}(\mathbf{x})=\bar{z} A(\mathbf{x})$	$A^{M}(\mathbf{x})=\bar{z} A(\mathbf{x})$
penev_B_ EQ0048 (5.30)	011	■ 1.000 ● N|+0	`A(\mathbf{x})=\mathcal{D}_{P^{M}}(\mathbf{x})`	$A(\mathbf{x})=\mathcal{D}_{P^{M}}(\mathbf{x})$	$A(\mathbf{x})=\mathcal{D}_{P^{M}}(\mathbf{x})$
penev_B_ EQ0049 (5.31)	011	■ 1.000 ● N|+1	`A\left(\mathbf{x}_{m}\right)=\left.0\right|_{m=1} ^{M-1} .`	$A\left(\mathbf{x}_{m}\right)=\left.0\right\|_{m=1}^{M-1} .$	$A(\mathbf{x}_{m})=\left.0\right\|_{m=1}^{M-1}.$
penev_B_ EQ0051 (5.33)	011	■ 1.000 ● N|+0	`A^{M}(\mathbf{x})=\frac{\mathcal{D}_{P M}(\mathbf{x})}{\mathcal{D}_{P M}\left(\mathbf{x}_{M}\right)}`	$A^{M}(\mathbf{x})=\frac{\mathcal{D}_{P M}(\mathbf{x})}{\mathcal{D}_{P M}\left(\mathbf{x}_{M}\right)}$	$A^{M}(\mathbf{x})=\frac{\mathcal{D}_{PM}(\mathbf{x})}{\mathcal{D}_{PM}(\mathbf{x}_{M})}$
penev_B_ EQ0052 (5.34)	011	■ 1.000 ● N|+1	`\Delta A^{m}(\mathbf{x})=A^{M}(\mathbf{x})\left[ \begin{array}{ll} -a^{m}\left(\mathbf{x}_{M}\right) & 1 \end{array}\right] .`	$\Delta A^{m}(\mathbf{x})=A^{M}(\mathbf{x})\left[\begin{array}{ll} -a^{m}\left(\mathbf{x}_{M}\right) & 1 \end{array}\right] .$	$\Delta A^{m}(\mathbf{x})=A^{M}(\mathbf{x})\left[\begin{array}{ll} -a^{m}(\mathbf{x}_{M}) & 1 \end{array}\right].$
penev_B_EQ0055	011	■ 1.000 ● N|+0	`120`	120	120
penev_B_ EQ0056 (5.37)	011	■ 1.000 ● W|+0	`\hat{f}_{n}=\sum_{m=1}^{M-1} \alpha_{m} f_{n-m}`	$\hat{f}_{n}=\sum_{m=1}^{M-1} \alpha_{m} f_{n-m}$	$\hat{f}_{n}=\sum_{m=1}^{M-1} \alpha_{m} f_{n-m}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Tables

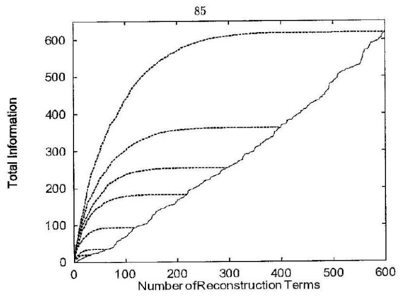
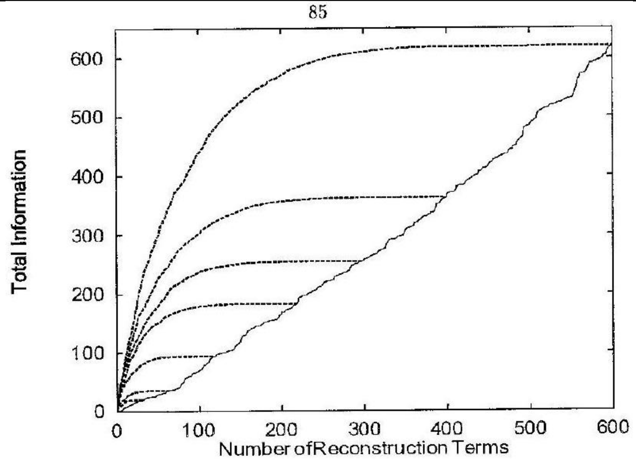
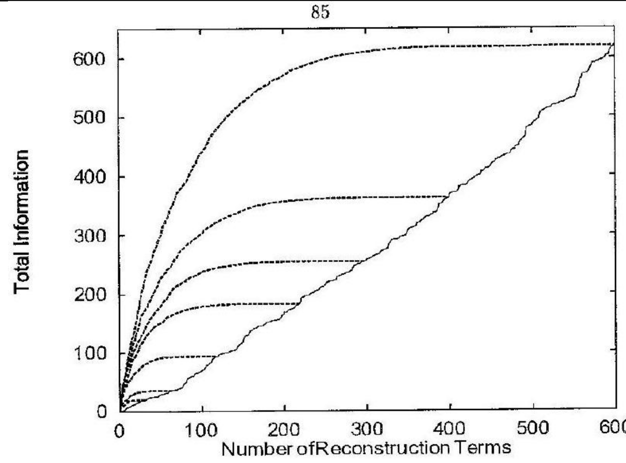
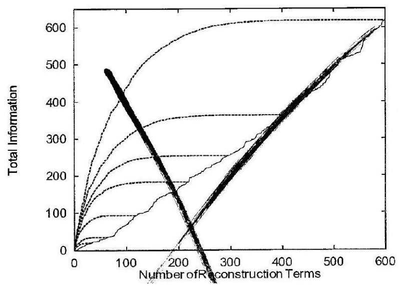
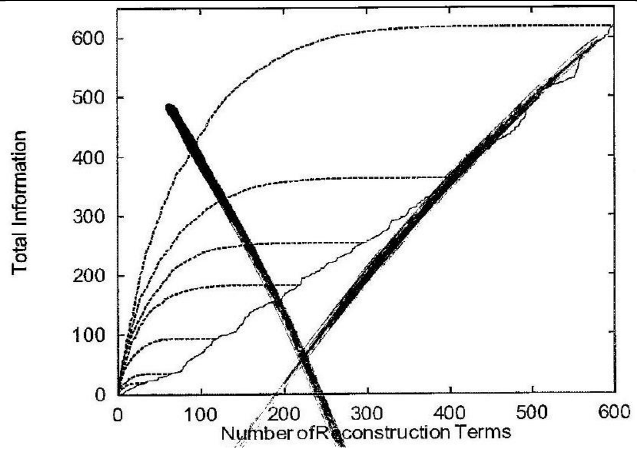
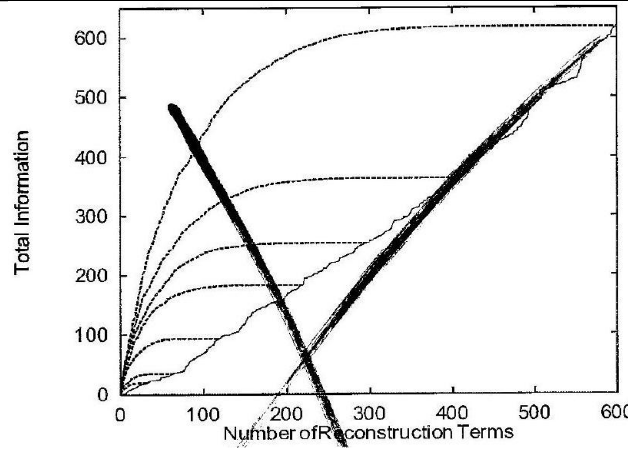
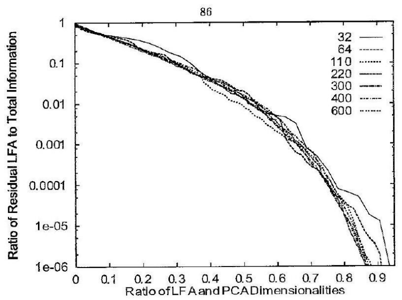
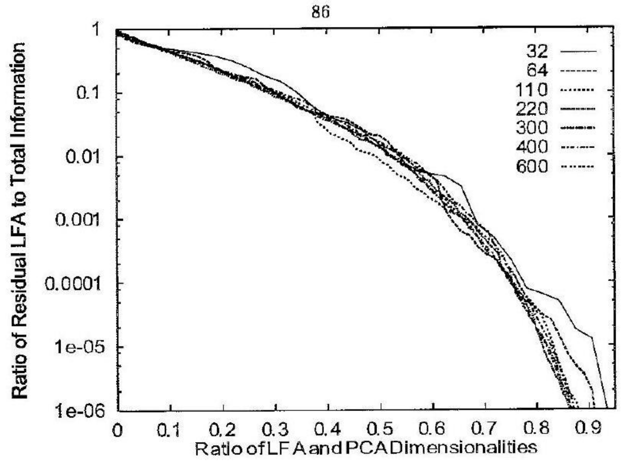
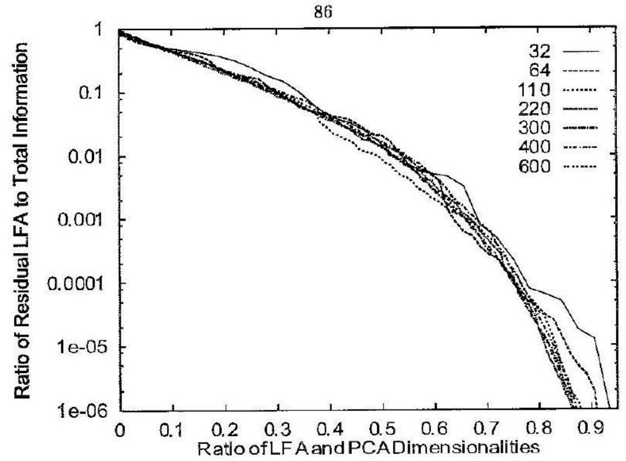
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
penev_B_TAB_001	000	—	`\begin{tabular}{||r|r|r|r||} \hline n & $s(n)$ & $ \frac{n(n-1)}{2}$ & $s(n) \times \frac{n(n-1)}{2}$ \\ \hline 1 & 26 & 0 & 0 \\ 2 & 769 & 1 & 769 \\ 3 & 145 & 3 & 435 \\ 4 & 143 & 6 & 858 \\ 5 & 12 & 10 & 120 \\ 6 & 17 & 15 & 255 \\ 7 & 8 & 21 & 168 \\ 8 & 3 & 28 & 84 \\ 9 & 2 & 36 & 72 \\ 10 & 3 & 45 & 135 \\ 11 & 16 & 55 & 880 \\ 12 & 1 & 67 & 67 \\ \hline & & & $N_{s}=3842$ \\ \hline \end{tabular}`	*(not rendered)*	—

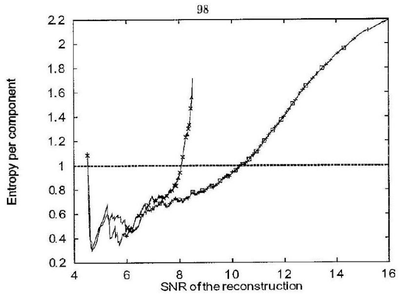
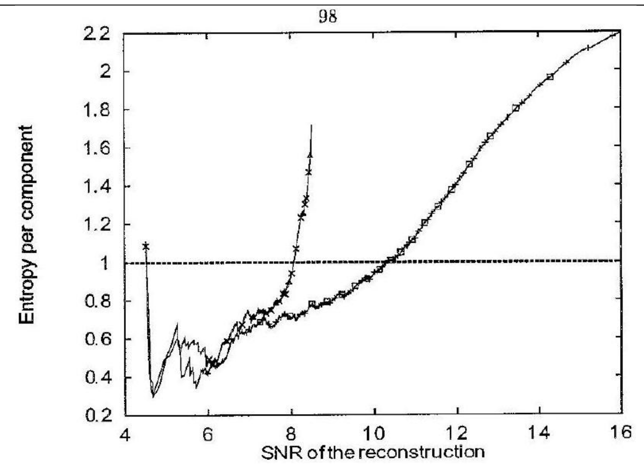
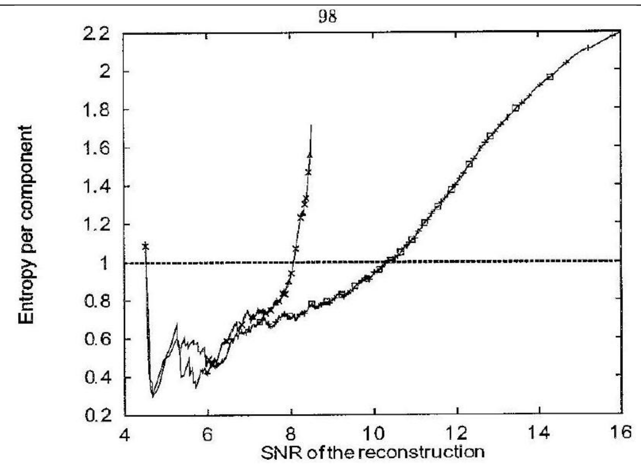
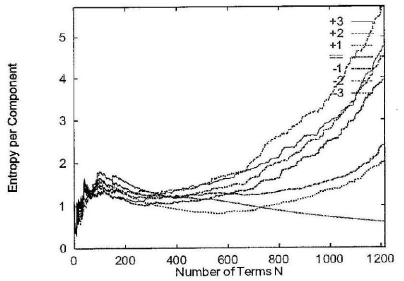
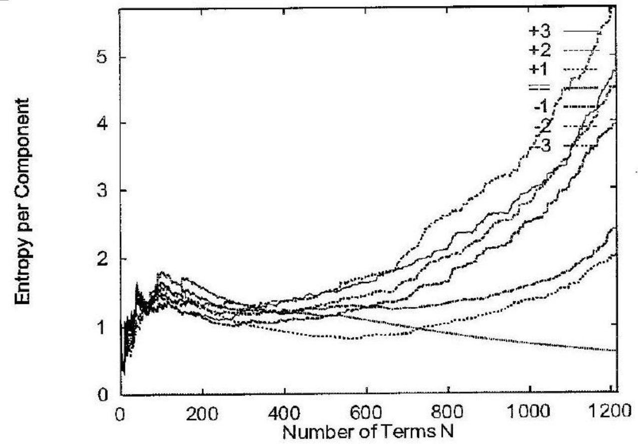
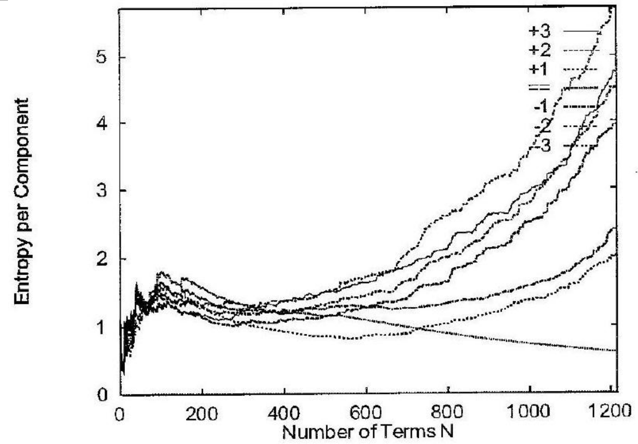
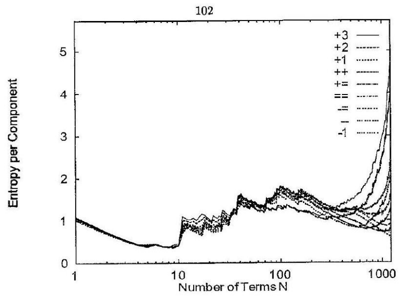
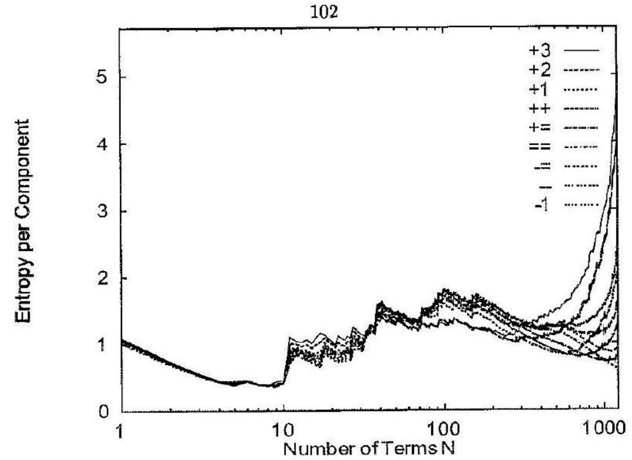
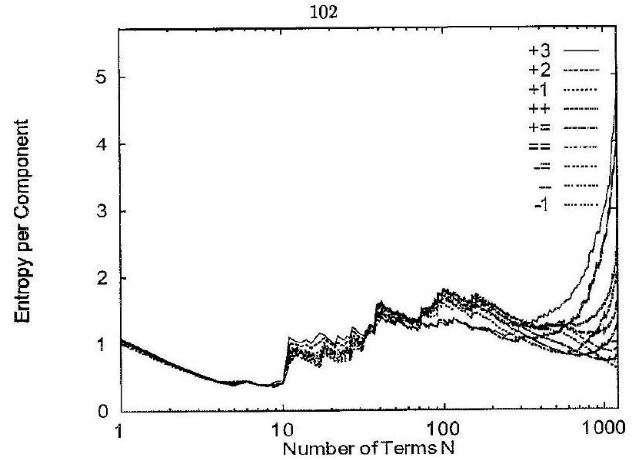
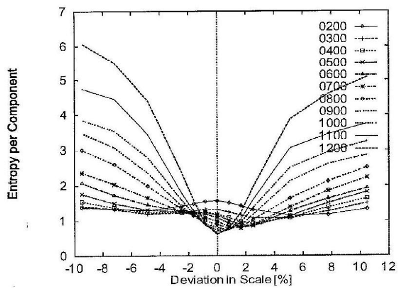
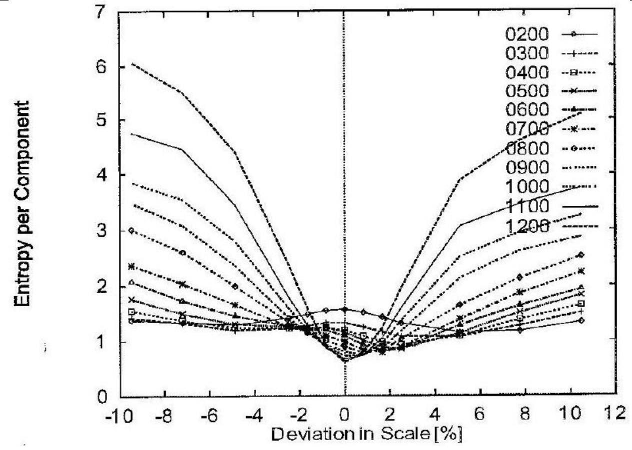
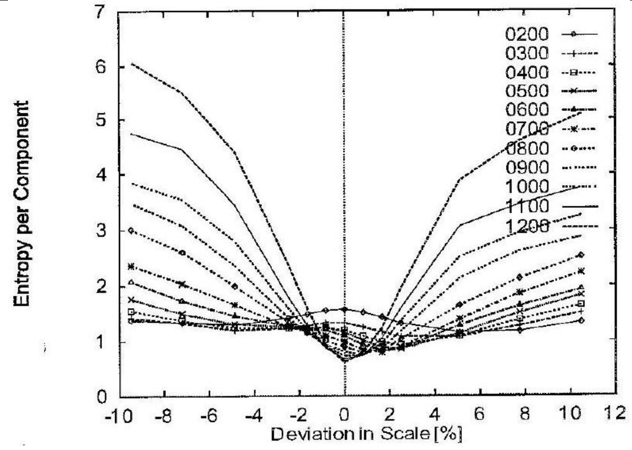
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

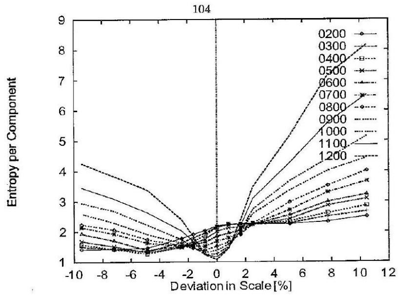
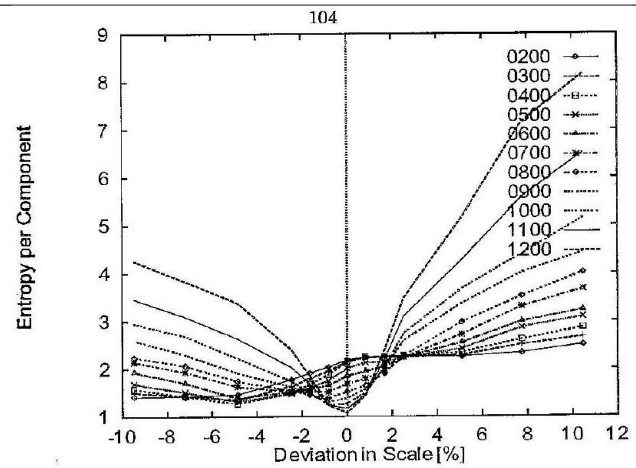
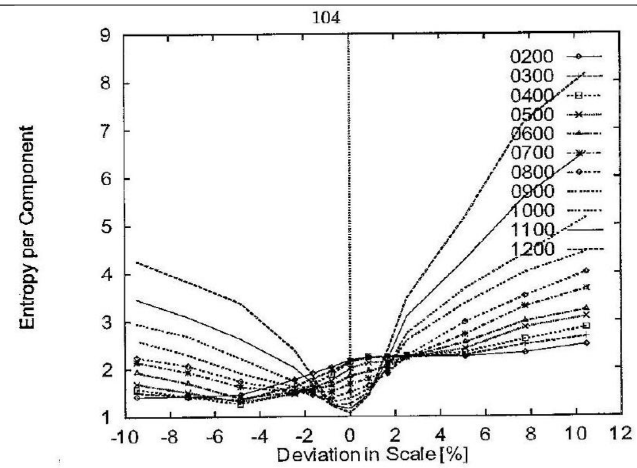
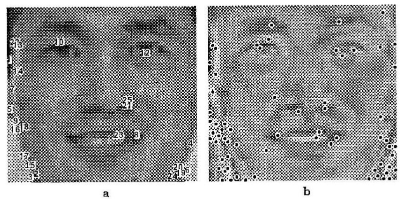
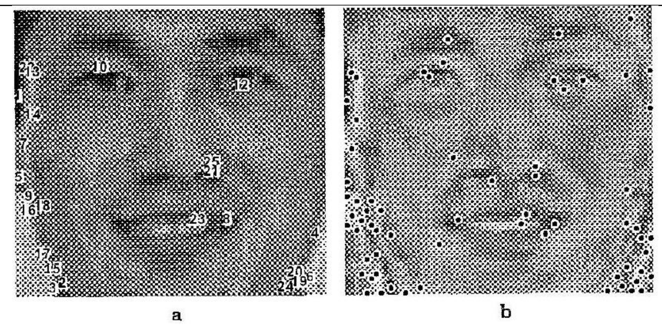
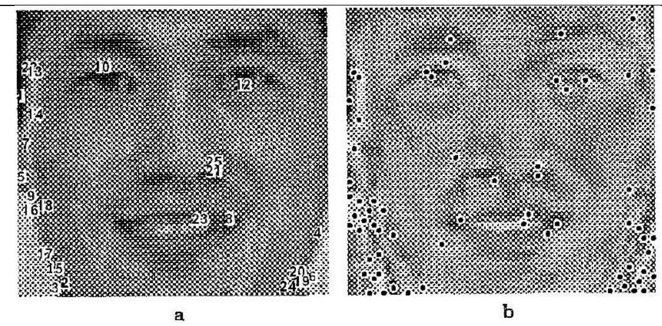
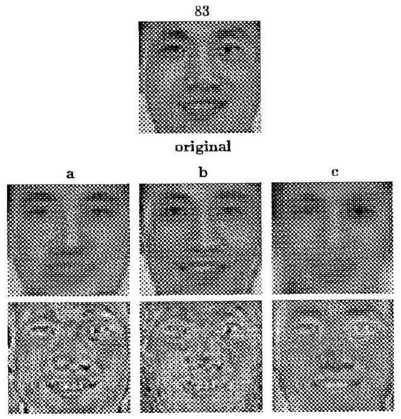
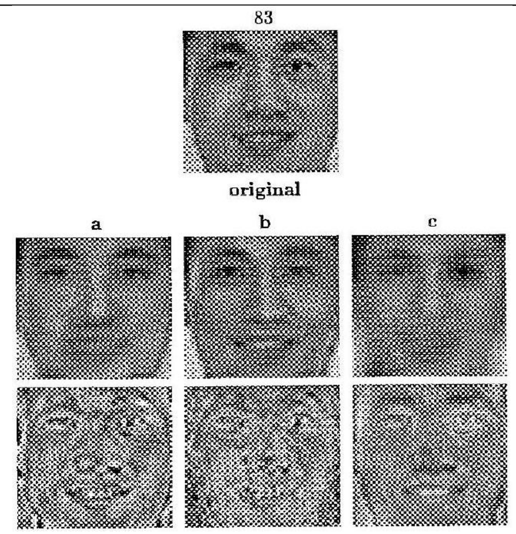
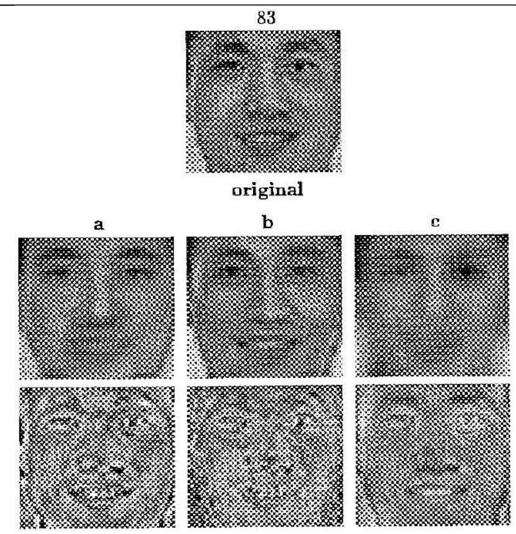
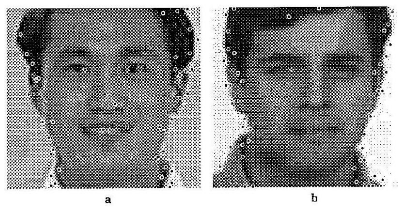
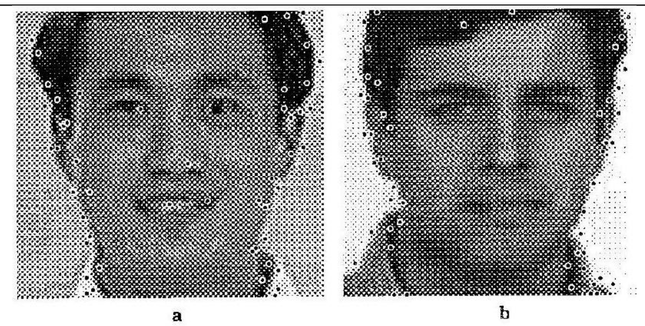
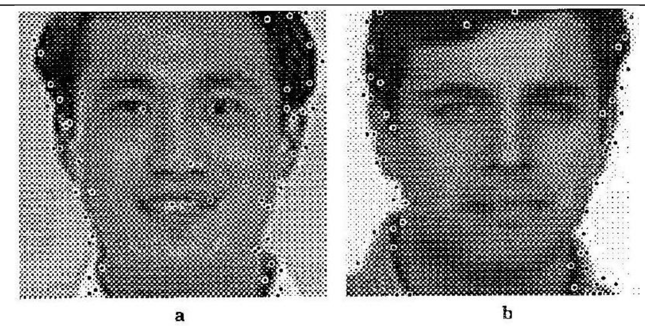
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

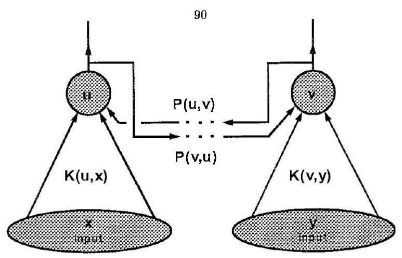
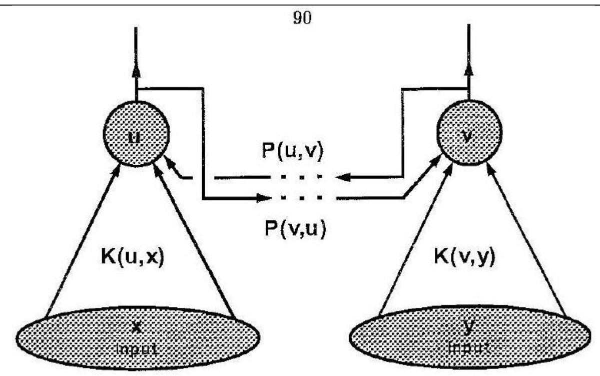
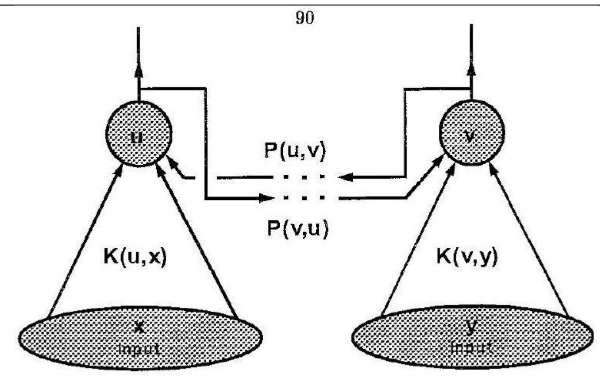
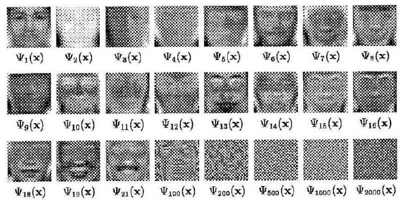
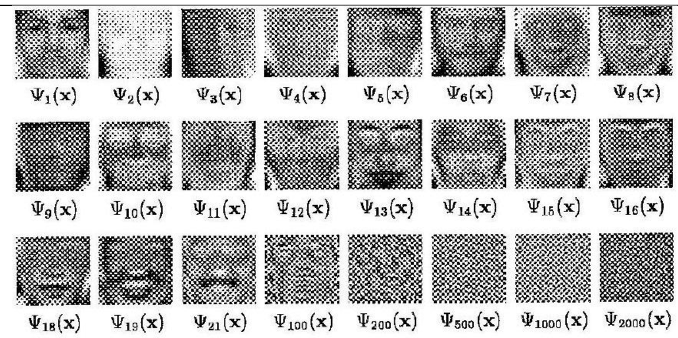
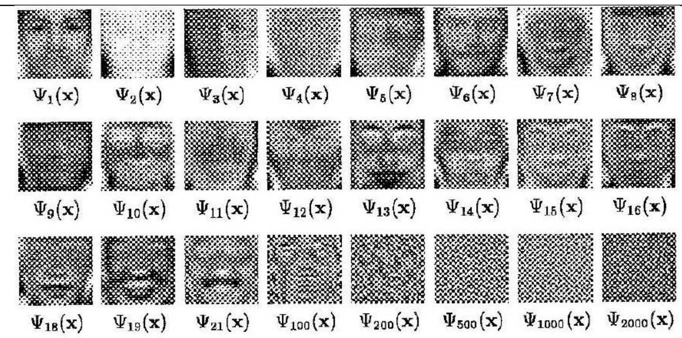
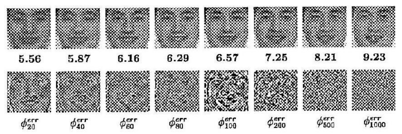
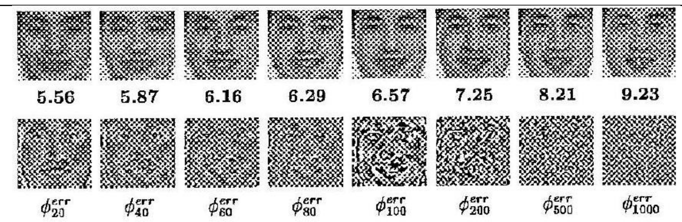
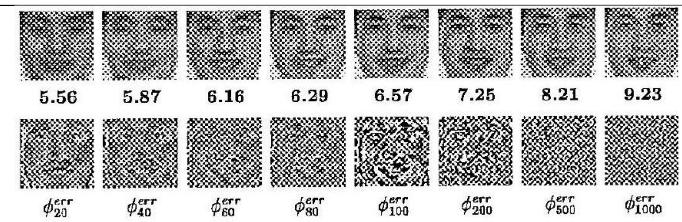
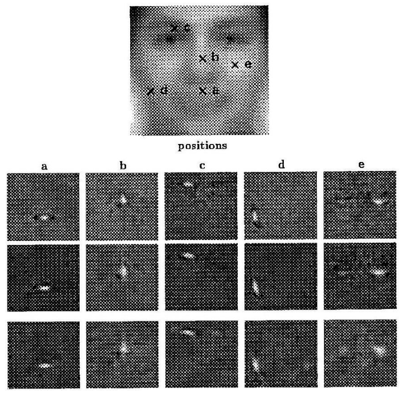
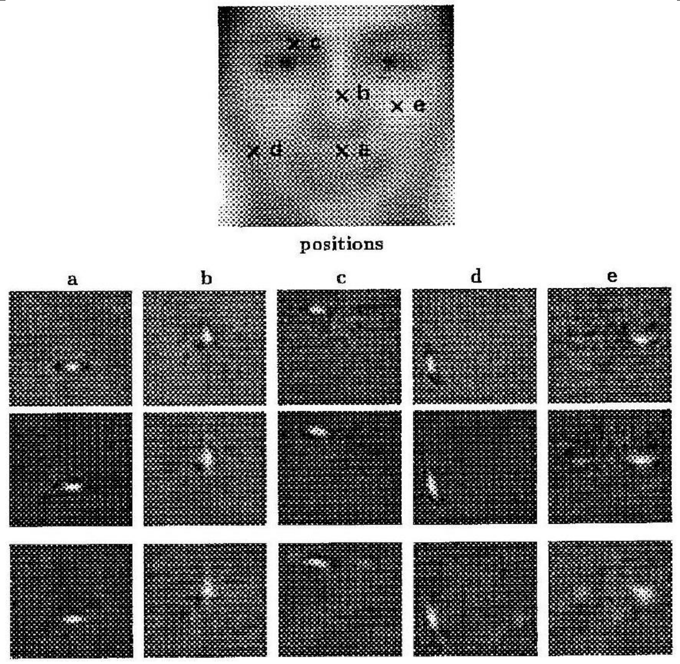
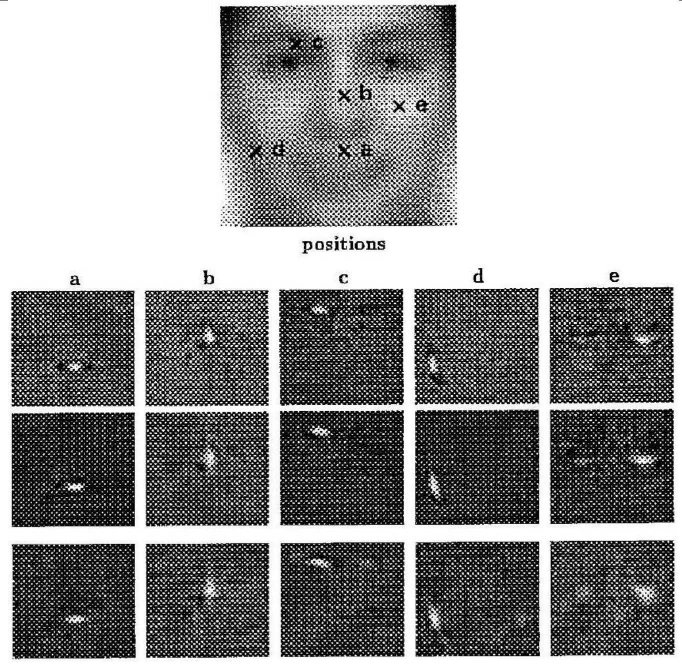
Image regions — rendered against scan

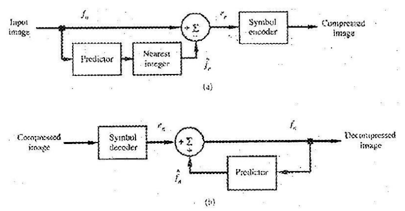
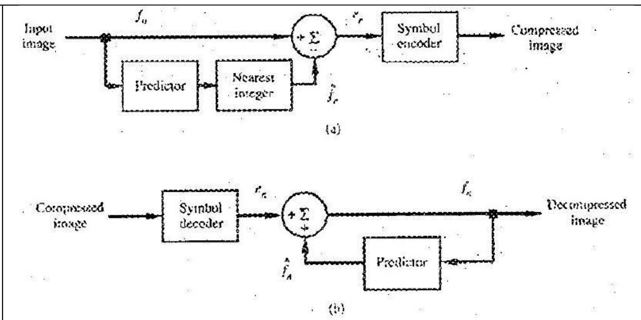
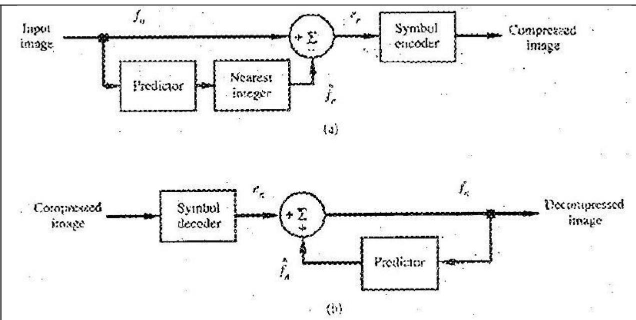
Two comparable cells per row. **Rendered**: the region's own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the `tex.zip` image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix's confidence, as the equation rows do. Setting a region's LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with `pdfdrill vision`; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
penev_B_PIC_0001	002	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-02_596_816_1430_1069.jpg			
penev_B_PIC_0002	003	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-03_580_820_103_130.jpg			
penev_B_PIC_0003	003	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-03_599_800_1439_137.jpg			

penev_B_PIC_0004	006	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-06_596_809_1437_139.jpg			
penev_B_PIC_0005	007	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-07_578_820_101_130.jpg			
penev_B_PIC_0006	007	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-07_601_809_1437_137.jpg			
penev_B_PIC_0007	007	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-07_580_809_91_1065.jpg			

penev_B_PIC_0008	007	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-07_601_809_1430_1074.jpg			
penev_B_DIA_0001	002	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-02_413_841_110_110.jpg			
penev_B_DIA_0002	002	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-02_685_651_1435_217.jpg			
penev_B_DIA_0003	003	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-03_418_813_94_1046.jpg			

penev_B_DIA_0004	004	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-04_480_758_1439_169.jpg			
penev_B_DIA_0005	005	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-05_432_866_94_1039.jpg			
penev_B_DIA_0006	006	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-06_281_870_94_103.jpg			
penev_B_DIA_0007	006	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-06_861_873_153_1037.jpg			

penev_B_DIA_0008	011	— *(dup)*	5fff204a-b675-4b41-925e-7cb5ba4e9fe4-11_411_800_128_1096.jpg			
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